Q1. A 2 cm long piece of silicon with cross-sectional area of 0.1 cm^2 is doped with donors at 10^{15} cm^{-3} is maintained at T=300K and has a resistance of 90 Ω. The saturation velocity of electrons in the silicon is 10^7 cm/sec for fields above 10^5 V/cm. **(5 Marks)**

a. Calculate the current if the applied voltage is 100V (2)

b. Calculate the current if the applied voltage of 10^6 V across it. (2)

c. In which case Ohm's law is valid? Explain briefly. (1)

Marking Scheme

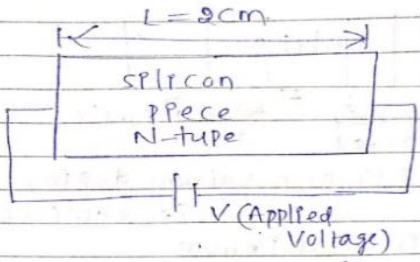
Part (a) award full marks for correct numerical value otherwise Give partial marks if correct formula is used.

Part (b) 1 mark for correct E field calculation and 1 mark for correct I value.

Part (c) award full marks for correct answer otherwise 0.

Solution

Q4



* The silicon piece is Maintained at T = 300 K.

Area (A) = 0.1 cm^2

$N_D = 10^{15} cm^{-3}$

Resistance (R) = 90 ohms

NOTE

$N_D - N_A >> 2n_i$

$\boxed{10^{15} >> 2\times10^{10}}$ → satisfied

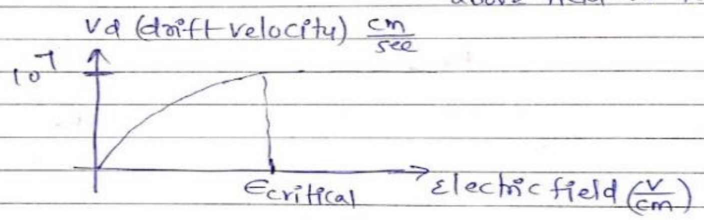
$(n_i)_{T=300K} = 10^{10}$

$n = N_D - N_A$

$n = N_D = 10^{15} cm^{-3}$

V_{sat} = saturation velocity

$V_{sat} = 10^7 \frac{cm}{sec}$ For above field 10^5 V/cm



Critical Electric field = 10^5 V/cm

$\mathcal{E} > \mathcal{E}_{critical} \Rightarrow \boxed{V_d = V_{sat} = 10^7 \frac{cm}{sec}}$

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Find Electron drift velocity

Resistance (R) = 90 ohms $\qquad$ Area = $0.1\,cm^2$

$R = \frac{\rho L}{A}$ ohms $\qquad$ $L = 2\,cm$

$\frac{RA}{L} = \rho$ ohm-cm

ρ = resistivity

$\rho = \frac{(90\,ohm)\,0.1\,cm^{\cancel{2}}}{2\,\cancel{cm}}$

$\rho = \frac{90 \times 0.1}{2} = \frac{9}{2}$ ohm-cm

$\boxed{\rho = 4.5\ ohm\text{-}cm}$

Resistivity.

Conductivity (J) = σE

$J = q\,n\,\mu_n\,\varepsilon$

ε = electric field

$\sigma = q\,n\,\mu_n$

$\rho = \frac{1}{q\,n\,\mu_n}$

$\mu_n = \frac{1}{q\,n\,\rho}$ $\qquad$ $\boxed{n = N_D \text{ For n-type material}}$

$\mu_n = \frac{1}{1.6 \times 10^{-19} \times 10^{15} \times 4.5}$

$\mu_n = \frac{1}{1.6 \times 4.5} \times 10^{4}\ \frac{cm^2}{V\text{-}sec}$

$\boxed{\mu_n = 1388.9\ \frac{cm^2}{V\text{-}sec}}$

applied voltage $(v) = 100$ volts

Length $(L) = 2\text{cm}$

$$\mathcal{E} = \frac{V}{L} = \frac{100\ V}{2\ cm} = 50\frac{V}{cm}$$

$$\boxed{\mathcal{E} < \mathcal{E}_{critical}} \Rightarrow \boxed{V_d = \mu_n \vec{\mathcal{E}}}$$

$$\boxed{\mathcal{E} > \mathcal{E}_{critical}} \Rightarrow \boxed{V_d = V_{d\,sat}}$$

$$\vec{V}_{d_n} = \mu_n \vec{\mathcal{E}}$$

$$V_{dn} = 1388.9 \times 50$$

$$\boxed{V_{dn} = 69.4444 \times 10^3 \frac{cm}{sec}}$$

Find the current through the piece when applied voltage is 10^6 volts

$V =$ Applied voltage

$V = 10^6$ volts

$L = 2\text{cm}$

$$\mathcal{E} = \frac{V}{L} = \frac{10^6}{2}\frac{V}{cm} = 5\times10^5\frac{V}{cm}$$

$$\boxed{\mathcal{E} > \mathcal{E}_{critical}} \quad \boxed{V_d = V_{d\,sat} = 10^7 \frac{cm}{sec}}$$

$J =$ current density

$J = \sigma \mathcal{E}$

$J = q n \mu_n \vec{\mathcal{E}}$

$J = q n\ V_{d\,sat}$

$J = 1.6\times10^{-19} \times 10^{15} \times 10^{7} = 1.6\times10^{-19}\times10^{22}$

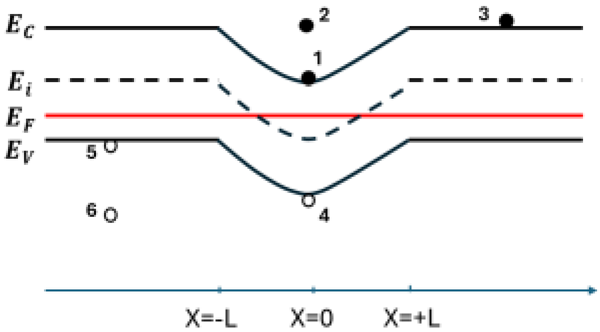
$\boxed{J = 1.6\times10^{3}\ A/cm^2}$ $\quad \boxed{\begin{array}{c} J = I/A \\ I = J\cdot A \end{array}}$

$I = 1.6\times10^3 \frac{A}{cm^2} \times 0.1 cm^2$

$\boxed{I = 160 \text{ Ampers}}$

- We can use Ohm's law in part (a) and get I = 1.11 A
- Ohm's law fails when Drift Velocity = Saturation Velocity, for Ohm's law to be valid, as E field increases, Vdrift should also change.

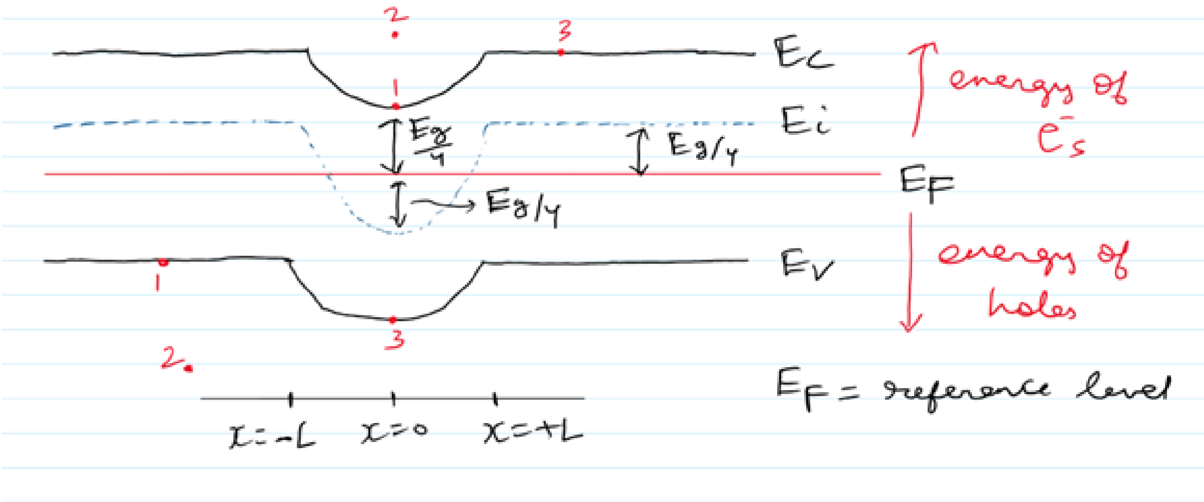
Q2. Consider the following energy band diagram. Take the semiconductor represented to be Si maintained at 300 K with $Ei - EF = Eg/4$ at x = ± L and $EF - Ei = Eg/4$ at x = 0. Note the choice of EF as the energy reference level. Find the K.E. and P.E. of the electrons and holes pictured on the diagram. **(3 Marks)**



Marking Scheme

Note: No partial marks. Every charge carrier should have the correct KE and PE. Each charge carrier (holes and electrons) carries 1 marks . 1/2 mark for correct K.E and 1/2 mark for correct P.E calculation.

Solution



for Kinetic energy (K.E.) $= E - E_c$

E: energy level of location of e^- sitting

for electron 1, we can visually see that it is sitting across the edge of E_c [or extremely near to E_c only, but let's assume that e^- 1 is at E_c energy level only]

So $K.E\, e^-|_1 = E_c - E_c = 0$

for electron 2, $(K.E.)_2 = E - E_c$

(the amount of band bending basically) for Conduction band

So from reference E_F, we can write (mathematically)

$$(K.E)_2 = (E_i - E_F)_{x=+L \text{ or } -L} + (E_F - E_i)_{\text{at } x=0}$$

$$(K.E.)_2 = \frac{E_g}{4} + \frac{E_g}{4} = \frac{1.12\text{ eV}}{2} = 0.56\text{ eV}$$

for electron 3 $(K.E.)_3 = E - E_c = E_c - E_c = 0$

[$\because$ e^- 3 is also sitting on edge of E_c so considering ~~its~~ it's energy level as E_c]

For Potential Energy : energy of holes increases downwards on the diagram or energy of e^-s increases upwards on diagram

$P.E. = E_c - E_F$ [$\because$ $E_c - E_{ref.}$] $\rightarrow$ FOR e^-s

for electron 1 [which is lying on conduction band]

$$(P.E)_1 = E_c - E_F = (E_c - E_i) - \underbrace{(E_F - E_i)}_{\text{at } x=0 \text{ origin}}$$

$$(P.E.)_1 = \frac{E_g}{2} - \frac{E_g}{4} = \frac{E_g}{4} = \frac{1.12}{4} = 0.28\text{ eV}$$

for electron 2 $P.E|_{2e^-} = E_c - E_F$ [P.E of e^- 1 & e^- 2 will be same]
↳ at same x-level (origin)

for electron 3

$$[P.E]_{3e^-} = E_c - E_F = (E_c - E_i) + (E_i - E_F) = \frac{E_g}{2} + \frac{E_g}{4}$$

$$= \frac{3E_g}{4} = \frac{3 \times 1.12}{4} = 0.84\,eV$$

for hole 1 $P.E|_{1hole} = E_V - E_{ref} = E_V - E_F$

$$P.E = (E_V - E_i) - (E_F - E_i) = \frac{E_g}{2} - \frac{E_g}{4} = \frac{E_g}{4} = 0.28\,eV$$

for hole 3 (at origin $x = 0$)

$$P.E = (E_V - E_i) + (E_i - E_F) = \frac{E_g}{2} + \frac{E_g}{4} = \frac{3E_g}{4} = 0.84\,eV$$

for hole 2 (sitting far away from V.B.)

but this can affect K.E but not P.E

So P.E of hole 1 & hole 2 will be same [at same x-level]

$$(P.E)_{1hole} = (P.E)_{2hole} = 0.28\,eV$$

Carrier	**K.E (eV)**	**P.E (eV)**
Electron 1	0	0.28
Electron 2	0.56	0.28
Electron 3	0	0.84
Hole 1	0	0.28
Hole 2	0.56	0.28

Hole 3	0	0.84

Q3. For a PN junction diode: **(6 Marks)**

a. Derive the expression for Built-in voltage (V_{bi}) (1)

b. Derive the expression for depletion width (W) (3)

c. Calculate V_{bi} and W if $N_A = 10^{19}\ cm^{-3}, N_D = 10^{17} cm^{-3}$ (2)

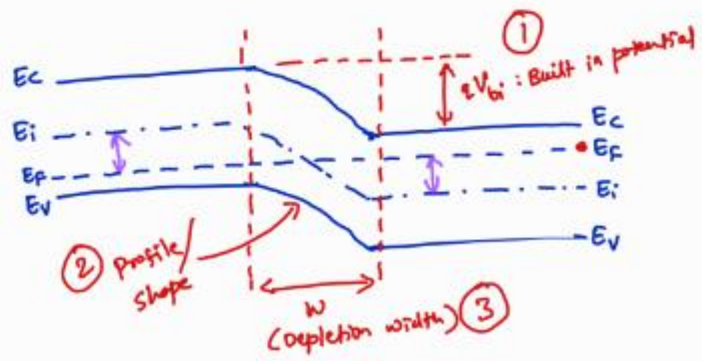
Marking Scheme

Part(a) complete derivation award 1 marks otheriwse half marks if close to the answer.

Part(b) Give 0.5 mark for Charge density, 1 mark for E-field derivation and 1 mark for final voltage derivation and 0.5 for final W derivation

Part(c) 1 mark for built in voltage and 1 mark for W calculation.

Solution



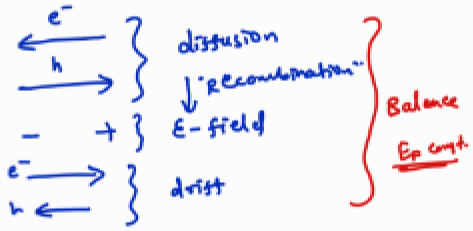
1) Built in potential

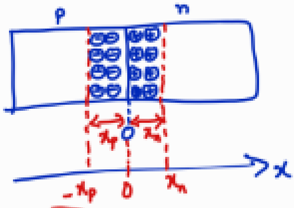
$$qV_{bi} = (E_i - E_F)_p + (E_F - E_i)_n$$

$$qV_{bi} = kT \ln(p_0/n_i) + kT \ln(n_0/n_i)$$

$10^{17} \times 10^{18}$

$$\Rightarrow \boxed{V_{bi} = \frac{kT}{q} \ln\left(\frac{p_0 n_0}{n_i^2}\right) = \frac{kT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right)}$$





Charge density

$-x_p \le x \le 0$

Exactly

$n_i e^{\frac{E_F - E_i(x)}{kT}}$

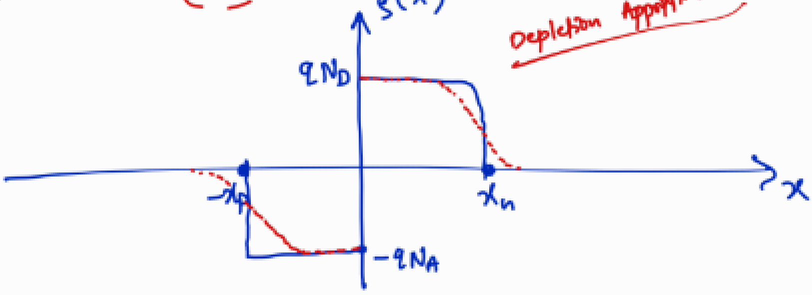
$-q\,(N_A(x) + p(x) - n(x))$

$$\rho(x) = -qN_A$$

$0 \le x \le x_n$

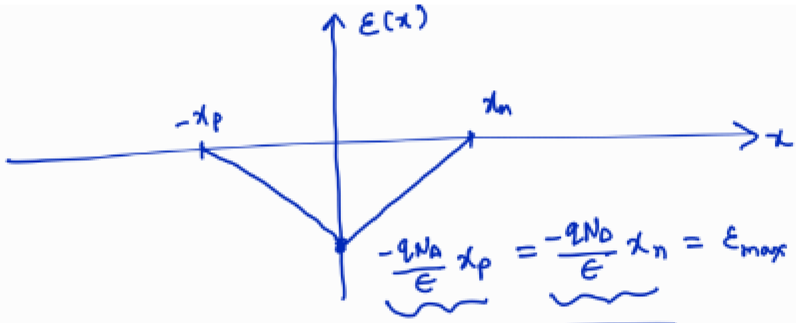
$$\rho(x) = qN_D$$

Depletion Approximation



② **$\mathcal{E}$-field** : $\dfrac{d\mathcal{E}(x)}{dx} = \dfrac{\rho(x)}{\epsilon}$

$\mathcal{E}(x) = \dfrac{-qN_A}{\epsilon}x + C_1$	$\mathcal{E}(x) = \dfrac{qN_D}{\epsilon}x + C_2$
@ $x = -x_p$; $\mathcal{E}(x) = 0$	@ $x = x_n$; $\mathcal{E}(x) = 0$
$C_1 = -\dfrac{qN_A}{\epsilon} \cdot x_p$	$C_2 = \dfrac{-qN_D}{\epsilon} x_n$
$\Rightarrow \mathcal{E}(x) = \dfrac{-qN_A}{\epsilon}(x + x_p)$	$\Rightarrow \mathcal{E}(x) = \dfrac{-qN_D}{\epsilon}(x_n - x)$



$$\underbrace{\frac{-qN_A}{\epsilon}}\, x_p = \underbrace{\frac{-qN_D}{\epsilon}}\, x_n = E_{max}$$

$$\boxed{N_A x_p = N_D x_n} \quad \text{—— (1)}$$

(3) Potential $\quad E(x) = \dfrac{-dV(x)}{dx}$

$$\Rightarrow E(x) = \frac{-qN_A}{\epsilon}(x + x_p) \qquad \Big| \qquad \Rightarrow E(x) = \frac{-qN_D}{\epsilon}(x_n - x)$$

$$V(x) = \frac{qN_A}{\epsilon}\left(\frac{x^2}{2} + x_p \cdot x\right) + C_3 \qquad \Big| \qquad V(x) = \frac{qN_D}{\epsilon}\left(x_n \cdot x - \frac{x^2}{2}\right) + C_4$$

LHS $\qquad\qquad\qquad$ RHS.

1) @ $x = -x_p$, $V(x) = 0$

2) @ $x = x_n$, $V(x) = V_{bi}$

3) @ $x = 0$, LHS = RHS

(1) $\qquad C_3 = \dfrac{qN_A}{\epsilon}\dfrac{x_p^2}{2}$,

(3) $\qquad C_3 = C_4$

(2) $$V_{bi} = \frac{qN_D}{\epsilon}\left(x_n^2 - \frac{x_n^2}{2}\right) + \frac{qN_A}{\epsilon}\frac{x_p^2}{2}$$

$$V_{bi} = \frac{qN_D}{\epsilon}\left(\frac{x_n^2}{2}\right) + \frac{qN_A}{\epsilon}\frac{x_p^2}{2}$$

from (1) $\quad N_A x_p = N_D x_n \Rightarrow x_p = \frac{N_D x_n}{N_A}$

$$\Rightarrow V_{bi} = \frac{qN_D}{2\epsilon} \cdot x_n^2 + \frac{q\cancel{N_A}}{2\epsilon}\frac{N_D^2 x_n^2}{N_A^{\cancel{2}}}$$

$$\Rightarrow V_{bi} = \frac{qN_D}{2\epsilon} x_n^2 + \frac{q}{2\epsilon} x_n^2 \cdot \frac{N_D^2}{N_A}$$

$$\Rightarrow V_{bi} = \frac{q x_n^2}{2\epsilon}\left(\frac{N_A N_D + N_D^2}{N_A}\right)$$

$$\Rightarrow V_{bi} = \frac{q x_n^2}{2\epsilon}\frac{N_D}{N_A}\left(N_A + N_D\right)$$

$$\boxed{x_n = \sqrt{\frac{2\epsilon . V_{bi}}{q}\frac{N_A}{N_D}\frac{1}{(N_A + N_D)}}}$$

$$\frac{KT}{q}\ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$x_p = \sqrt{\frac{2\epsilon V_{bi}}{q} \frac{N_D}{N_A} \frac{1}{(N_A + N_D)}}$$

$$W = x_p + x_n$$

$$\boxed{W = \sqrt{\frac{2\epsilon}{q} V_{bi} \left(\frac{1}{N_A} + \frac{1}{N_D}\right)}}$$

V_{bi} = 0.916 V and **W** = 0.108 µm

Q4. For a PN junction diode: **(6 Marks)**

a. Derive the ideal PN junction diode current equation and plot the I-V in log and linear scale (4)

b. Calculate current at room temperature if saturation current $I_0 = 1nA$ and applied voltage is

1. Voltage is -5V
2. Voltage is 0.2V
3. Voltage is 1 V
4. Voltage is 100V

Is this calculated value of current reasonable for all the voltage values? Explain briefly. (2)

Marking Scheme

Part (a) derivation 3 marks and plots in log and linear scale 0.5 mark each

Part (b) half mark for each subpart in this question

Solution:-

A) Ideal Diode Current Derivation

$$J_{e\text{-}diff} = qD_n \frac{dn}{dx}$$

$$= qD_n \frac{\Delta n}{L_n}$$

$$= \frac{qD_n}{L_n}\left(n_p - n_{po}\right)$$

$$J_{e\text{-}diff} = \frac{qD_n}{L_n}\left(n_{po}\, e^{V_a/V_T} - n_{po}\right)$$

$$J_{e\text{-}diff} = \frac{qD_n}{L_n} \cdot n_{po}\left(e^{V_a/V_T} - 1\right)$$

$$J_{h\text{-}diff} = \frac{qD_p}{L_p} \cdot p_{no}\left(e^{V_a/V_T} - 1\right)$$

$$J_T = J_{e\text{-}diff} + J_{h\text{-}diff}$$

$$J_T = \underbrace{q\left(\frac{D_n}{L_n} n_{po} + \frac{D_p}{L_p} p_{no}\right)}_{\text{saturation current } (nA,\ pA,\ fA)}\left(e^{V_a/V_T} - 1\right)$$

$$\boxed{\begin{aligned} J_T &= J_0 \cdot \left(e^{V_a/V_T} - 1\right) \\ I &= I_0\left(e^{V_a/V_T} - 1\right) \end{aligned}}$$

$\underline{V > 0}$

$$J_T = J_0\, e^{V_a/V_T}$$

Electron Diffusion Current Density ($J_{e\text{-}diff}$)

$$J_{e\text{-}diff} = qD_n \frac{dn_p}{dx}$$

$$J_{e\text{-}diff} = qD_n \frac{\Delta n_p}{L_n}$$
$$J_{e\text{-}diff} = \frac{qD_n}{L_n}(n_p - n_{p0})$$

Applying the law of the junction

$$n_p = n_{p0}e^{V_a/V_T}$$
$$J_{e\text{-}diff} = \frac{qD_n}{L_n}(n_{p0}e^{V_a/V_T} - n_{p0})$$
$$\boxed{J_{e\text{-}diff} = \frac{qD_n}{L_n}n_{p0}(e^{V_a/V_T} - 1)}$$

Hole Diffusion Current Density ($J_{h\text{-}diff}$)

$$J_{h\text{-}diff} = \frac{qD_p}{L_p}p_{n0}(e^{V_a/V_T} - 1)$$

Total Current Density (J_T)

$$J_T = J_{e\text{-}diff} + J_{h\text{-}diff}$$
$$J_T = q\left(\frac{D_n}{L_n}n_{p0} + \frac{D_p}{L_p}p_{n0}\right)(e^{V_a/V_T} - 1)$$

Ideal Diode Equation

Saturation current density:

$$J_0 = q\left(\frac{D_n}{L_n}n_{p0} + \frac{D_p}{L_p}p_{n0}\right)$$

Multiplying by area A:

$$\boxed{I = I_0(e^{V_a/V_T} - 1)}$$

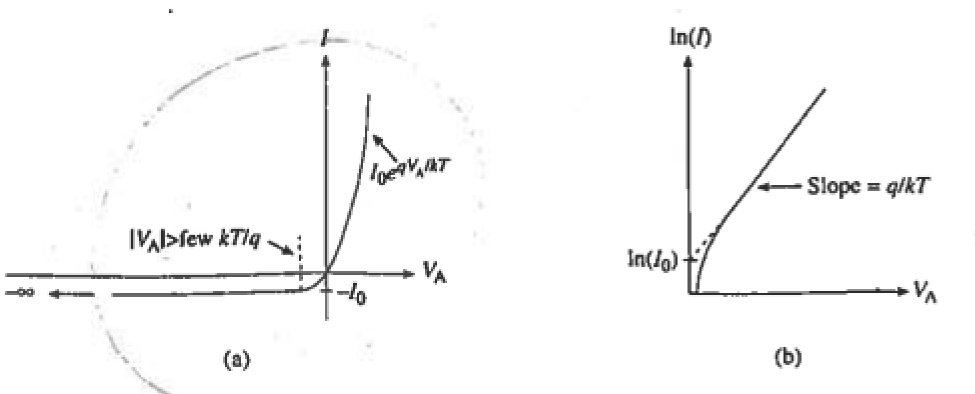


Figure 6.6 Ideal diode I–V characteristics: (a) linear plot identifying major features; (b) forward-bias semilog plot.

b)

I_0=1nA=10^{-9}A

Room Temperature:

$$V_T = 0.0259\text{ V}$$

Diode equation:

$$I = I_0\left(e^{V_a/V_T} - 1\right)$$

Applied Voltage (V_a)	Calculation	Resulting Current (I)
−5 V	$1\ \text{nA}\left(e^{-5/0.0259} - 1\right)$	≈ -1 nA(Reverse saturation)
0.2 V	$1\ \text{nA}\left(e^{0.2/0.0259} - 1\right)$	$\approx 2.25\ \mu$A
1 V	$1\ \text{nA}\left(e^{1/0.0259} - 1\right)$	$\approx$ 58.6 MA(Non-physical)
100 V	$1\ \text{nA}\left(e^{100/0.0259} - 1\right)$	Extremely large (**Impossible**)

Q5. For a NPN structure shown below, assume the P-region length (L_P) is sufficiently large so that the depletion regions do not merge from two junctions. Assume Depletion approximation. **(20 Marks)**

N	P	N
$N_D = 10^{20}/cm^3$	$N_A = 10^{18}/cm^3$	$N_D = 10^{20}/cm^3$
	$\longleftrightarrow$ L_P	

---

a. Calculate the potential barrier value **(1 Marks)**

Marking Scheme **Award 1 marks for correct value otherwise 0.**

$$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$V_{bi} = 1.03V$

b. Calculate depletion width (in nm) in N-region and P-region. Which region has larger portion of total depletion width? **(2 Marks)**

Marking Scheme **If the steps are correct, then deduct only 0.5 marks Here Total is 1.5 marks . ½ mark for correct answer.**

Solution

On either side of the junction depletion widths will be same

According to the formula, we get

$$x_N = \sqrt{\frac{2\varepsilon_s V_{bi}}{q} \cdot \frac{N_A}{N_D(N_A + N_D)}}$$

x_n = 0.37 nm

$$x_P = x_N \frac{N_D}{N_A}$$

x_p = 36 nm

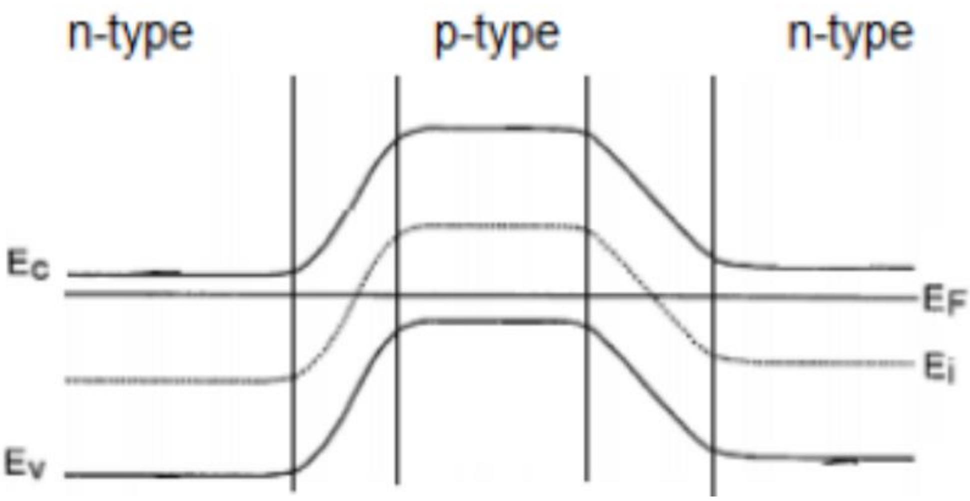
W = xp + xn = 36.36 nm

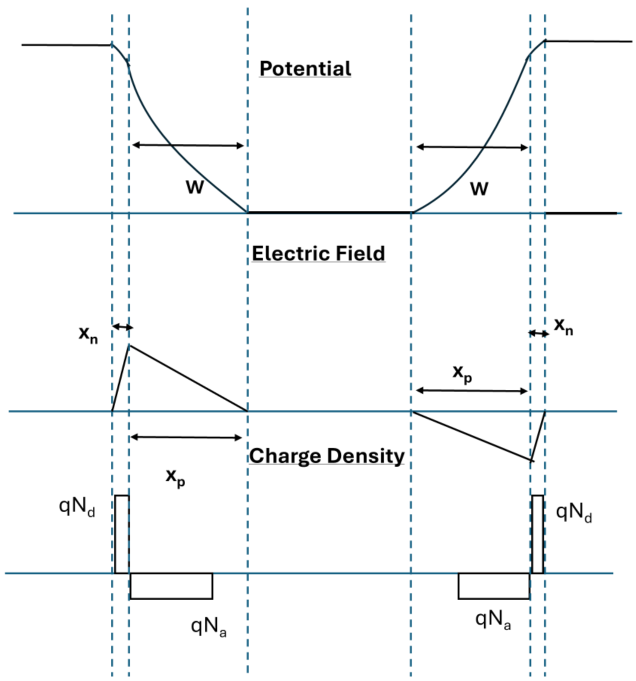
Since Nd >> Na almost all the depletion width will be inside P region. Since there are two junctions we will have Total Depletion width = 2 W = 72 nm

c. Draw Band diagram, potential, E-field and charge density profile. **(4 Marks)**

Marking Scheme **Deduct ½ mark for each mistake otherwise award full marks.**

Band Diagram

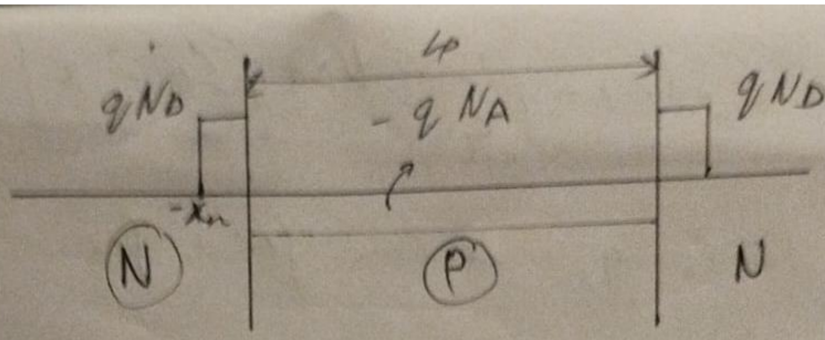




- Due to different depletion widths we can observe that the profiles on P and N sides differ.

d. Derive the equation for the new potential barrier and show its dependence on N_A, L_P **(3)**

Marking Scheme **Give 1 mark for Charge density, 1 mark for E-field derivation and 1 mark for final voltage derivation.**



If we balance the charges, we get

$$x_n q N_D = \frac{q N_A L_p}{2}$$

$$\Rightarrow 2x_n N_D = N_A L_p \text{ or } \boxed{N_D x_n = \frac{N_A L_p}{2}} \quad (1)$$

W.K.T, $\dfrac{\partial E}{\partial(x)} = \dfrac{\rho(x)}{\varepsilon_0}$

Taking $\vec{E}$ field on N side, we get

$$E_1(x) = \frac{1}{\varepsilon} \int_{-x_n}^{x} \rho(x)\, dx$$

$$\Rightarrow E_1(x) = \frac{1}{\varepsilon} \int_{-x_n}^{x} q N_D$$

$$\Rightarrow E_1(x) = \frac{q N_D}{\varepsilon} [x + x_n]$$

For E_{max}, we take $x = 0$

$$\Rightarrow E_1(max) = \frac{q N_D}{\varepsilon} x_n$$

using (1), we get

$$E_1(max) = \frac{q N_A L_p}{\varepsilon \times 2}$$

$$E_2(x) = \frac{1}{\epsilon} \int_{x}^{\frac{L_p}{2}} \rho(x)\, dx$$

$$\Rightarrow E_2(x) = \left(\frac{1}{\epsilon} q N_A\right)\left[\frac{L_p}{2} - x\right]$$

Similarly, we can find for NP junction;

$$V(x) = -\int_{-x_n}^{x} E(x)\, dx$$

$$V_b = \frac{1}{2} E_{max} \frac{L_p}{2}$$

$$V_b = \frac{q\, N_A\, l_p^2}{8\, \varepsilon_s}$$

e.Calculate the new potential barrier value. **(2 marks)**

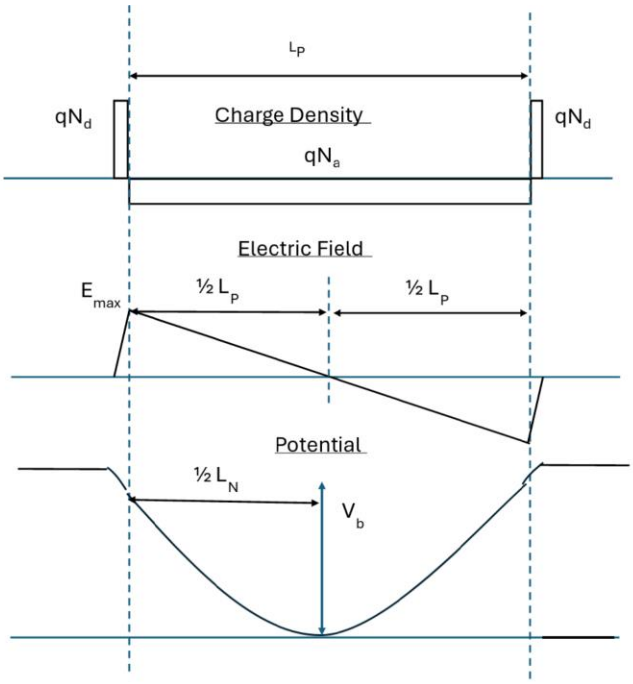
Marking Scheme If the Steps and final answer is wrong deduct only 1.2 marks.

Solution

New potential Barrier value is 0.17 V

f.Draw the Band diagram, potential, E-field and charge profile when $L_P = 30nm$. Calculate and annotate peak barrier value, peak electric field values on the plots? **(4 marks)**

Marking Scheme **1 mark each for each diagrams (Total 3 marks) and 1 mark for annotation.**



❖ Notice the depletion regions have merged and we get a single box like charge density on p side.

g. With $L_P = 30nm$, to achieve the peak potential barrier of 0.5V, what should be N_A? **(2 Marks)**

Marking Scheme **If the Steps and final answer is wrong deduct only 1.2 marks.**

$N_a = 2.87 \times 10^{19} /cm^3$

h. With $N_A = 10^{18}/cm^3$, to achieve the peak potential barrier of 0.5V, what should be the L_P in nm? **(2 Marks)**

Marking Scheme **If the Steps and final answer is wrong deduct only 1.2 marks.**

$L_P = 160$ nm